

Predictive Statistical Problems Assignment Part 2

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Air Cargo Freight

Group 10

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(DAMO-510-4) Predictive Analytics

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Predictive Statistical Problems Assignment Part 2

Statement of Purpose

The main idea of this project is to use predictive analytics methods to air cargo freight data, and to analyze and predict deviation between planned and actual transport process times. Proper forecasting of delays in freight handling is essential towards improving reliability of supply chain, reducing cost of operations as well as increasing customer satisfaction. Using statistics and machine learning models, this project will identify systematic inefficiencies and give practical information about logistics management.

Scope of the Project

In this project, the analysis is limited to the given air cargo freight dataset that annotates planned and effective durations of several transport milestones of incoming and outgoing legs. The project has the following major tasks.

- Pretreatment of data and cleansing
- Exploratory data analysis (EDA)
- Delay calculations and leg/hop analysis as feature engineering.
- Planned versus actual duration- stats testing.
- Regression and classification based predictive modeling.
- Model evaluation, interpretation and development of business recommendations.

External market disruptions, weather conditions, and cost related information are not factors that fall under the scope of the dataset since these are outside the dataset.

Predictive Statistical Problems Assignment Part 2

Background Research and Literature Review

Forecasting delays and optimizing supply chain operations Predictive analytics have found more use in logistics. Gharehgozli, Roy, and Pedraza-Martinez (2019) proved the importance of predictive models in port and freight operations and the importance of machine learning in predicting delays in the work process. On the same note, Kourounioti, Mourtzis, and Doukas (2020) focused on the application of statistical and data-driven techniques to make logistics networks more resilient. These papers give theoretical basis of the application of regression and classification models to predict deviations in cargo process timelines.

References

- Gharehgozli, A. H., Roy, D., & Pedraza-Martinez, A. J. (2019). Predictive analytics in logistics: A review. *International Journal of Production Research*, 57(15-16), 4806–4823.
- Kourounioti, I., Mourtzis, D., & Doukas, M. (2020). Big data analytics for resilient logistics and supply chain management. *Procedia Manufacturing*, 51, 1632–1639.

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Design and Data Collection Methods

The data set is based on a freight process log of an airport cargo and has planned and measured durations of primary milestones, such as check-in (RCS), departure (DEP), arrival (RCF), and delivery (DLV). An instance of a process can has a combination of three incoming legs and one outgoing leg, and the airports and hops are identified.

Variable	Description
Nr	Unique process ID
Legs	Number of incoming legs (1–3)
i1_rcs_p/e	Planned/Effective duration of incoming leg 1 check-in
i1_dep_1_p/e	Planned/Effective duration of departure segment 1 (leg 1)
i1_rcf_1_p/e	Planned/Effective duration of arrival segment 1 (leg 1)
i1_dlv_p/e	Planned/Effective duration of delivery (leg 1)
i2_rcs_p/e	Planned/Effective duration of incoming leg 2 check-in
i3_rcs_p/e	Planned/Effective duration of incoming leg 3 check-in
o_rcs_p/e	Planned/Effective duration of outgoing leg check-in

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o_dep_1_p/e	Planned/Effective duration of departure segment 1
o_rcf_1_p/e	Planned/Effective duration of arrival segment 1
o_dlv_p/e	Planned/Effective duration of delivery

Internal freight logs provided data which was standardized to be analyzed. Each milestone was also mined and reproduced planned and actual durations to be used in the statistical and predictive modeling.

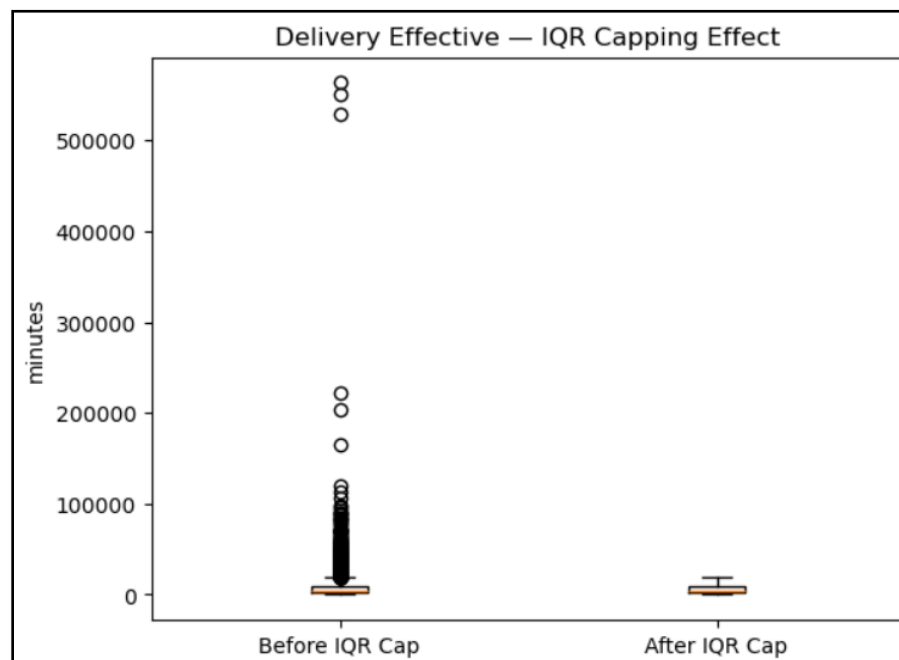
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Methodology / Strategies

Data Preprocessing

- The missing values were validated and imputed by statistical means.
- Boxplot and statistical thresholds were used to detect outliers which were appropriately dealt with.
- All milestones used new features, e.g. $\text{delay} = \text{effective} - \text{planned}$, which were engineered.

Figure 1. Boxplot of Planned vs Effective Times (Outlier Detection)



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Exploratory Data Analysis (EDA)

- Effective durations had heavier tails, and planned durations had right-skewness.
- Correlation analysis indicated the presence of high degrees of relationship between planned times and effective times under the same milestone.

Figure 2. Histogram of Planned vs Effective Durations

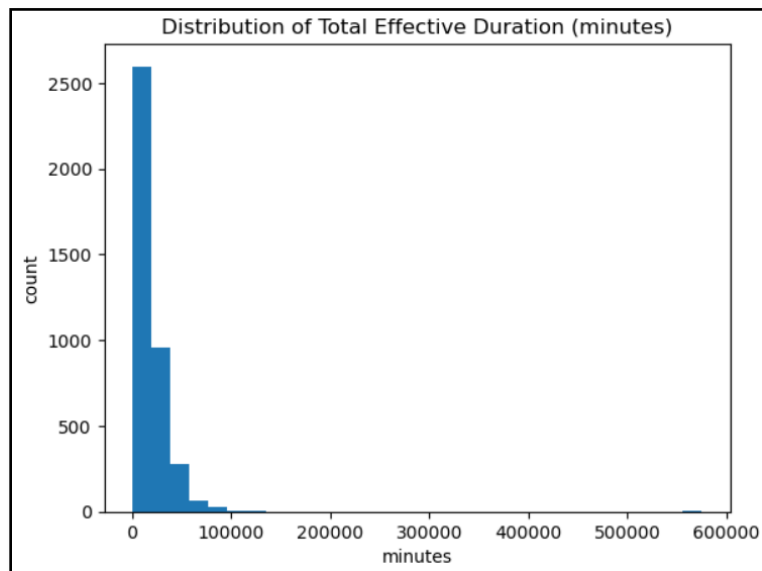
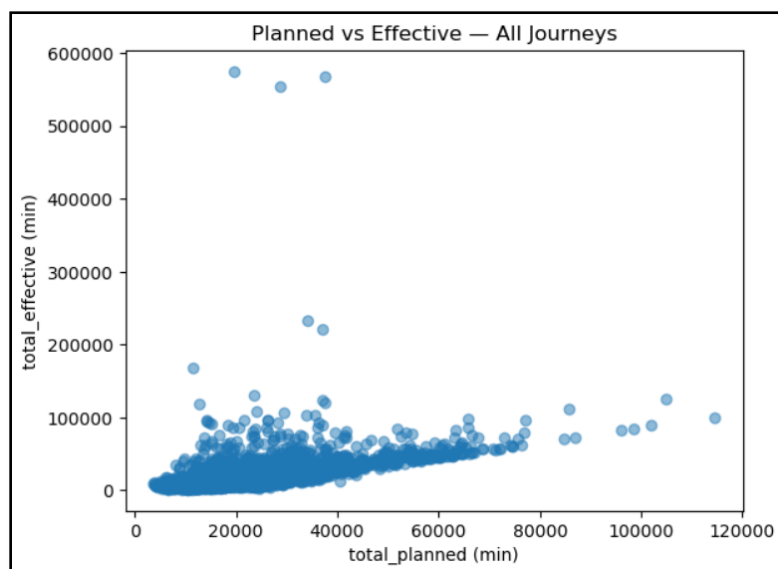


Figure 3. Planned vs Effective (All Journeys)



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Figure 4. Total Effective by Number of legs

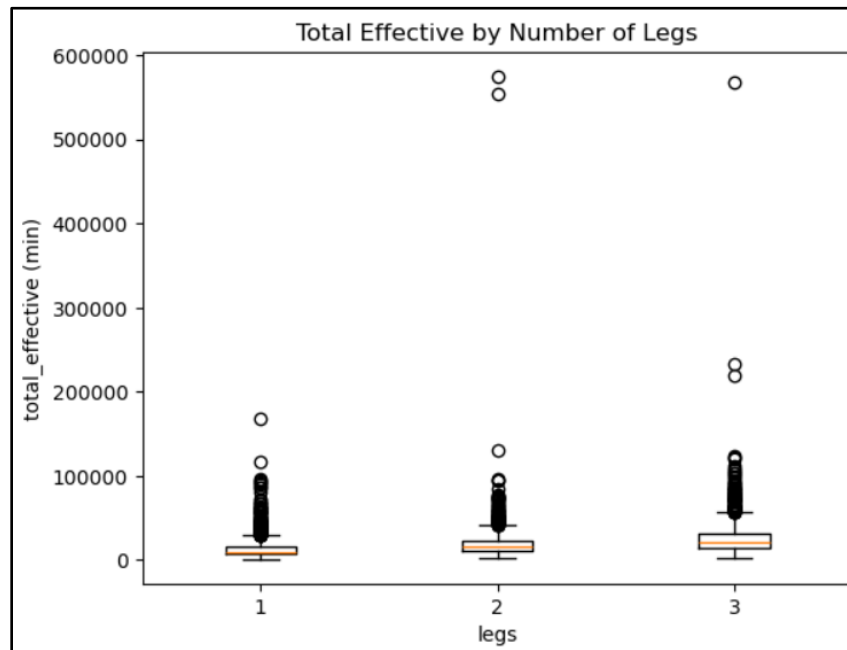
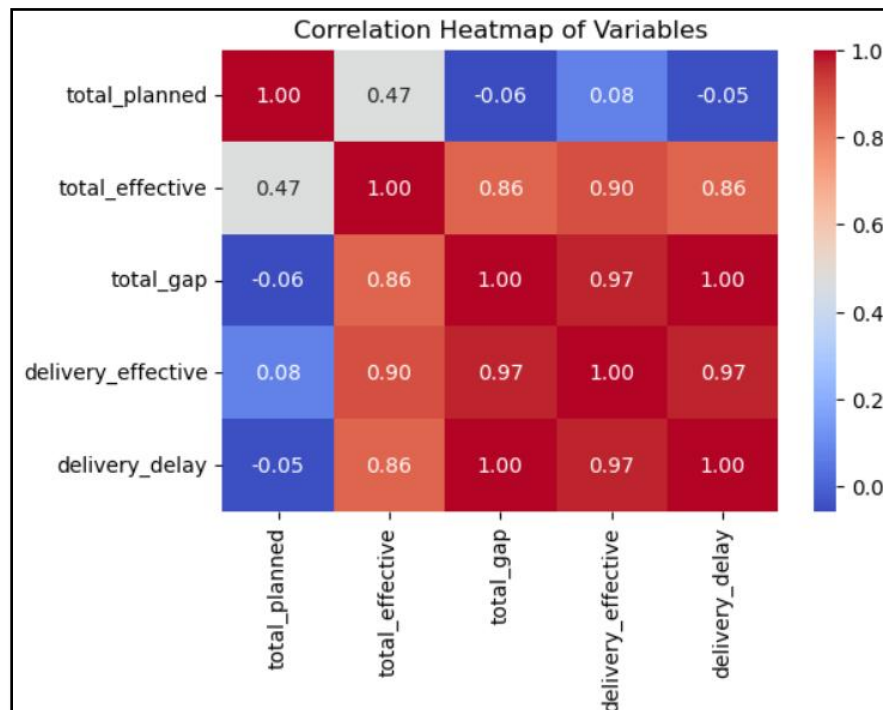


Figure 5. Correlation Heatmap of Variables



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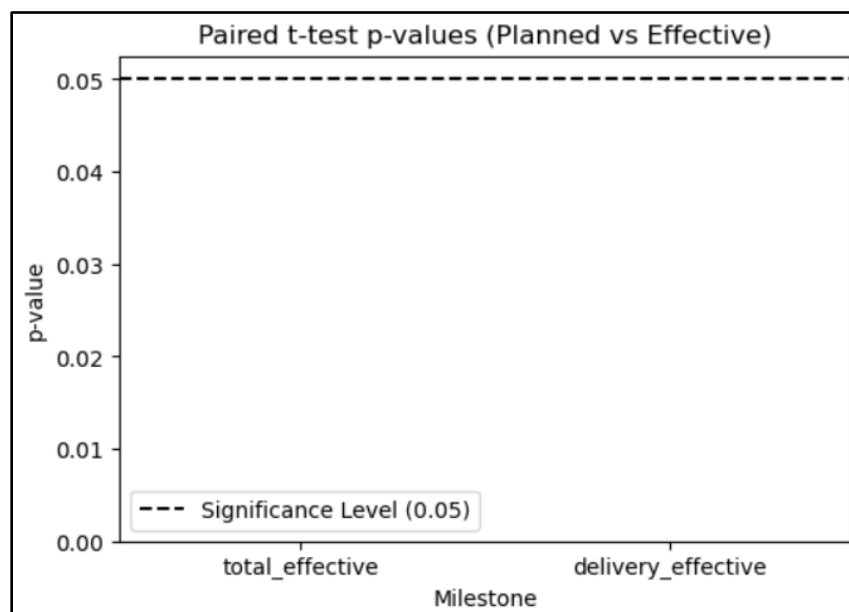
Statistical Testing

The comparison of planned and actual duration was done using paired t-tests. Most of the milestones were statistically significantly delayed, which proved systematic inefficiencies.

Figure 6. Paired t-test Statistics (Planned vs Effective)



Figure 7. Paired t-test p-values



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Predictive Modelling

- Prediction of effective times was done using linear regression of planned times, with moderate to strong explanatory power.
- Random Forest regression enhanced accuracy and minimized the mean absolute error (MAE).

Figure 8. Residuals vs Predicted

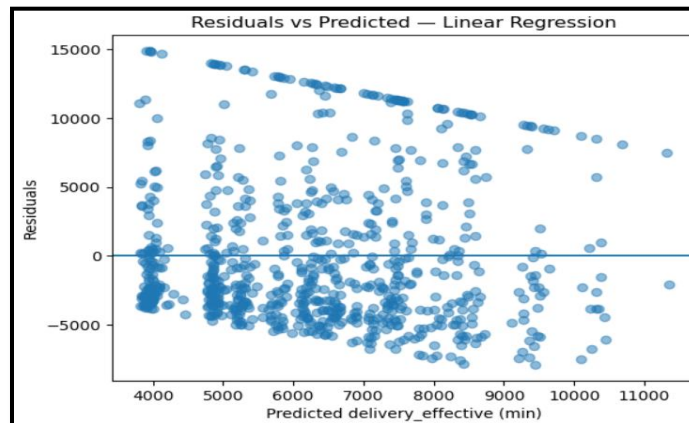
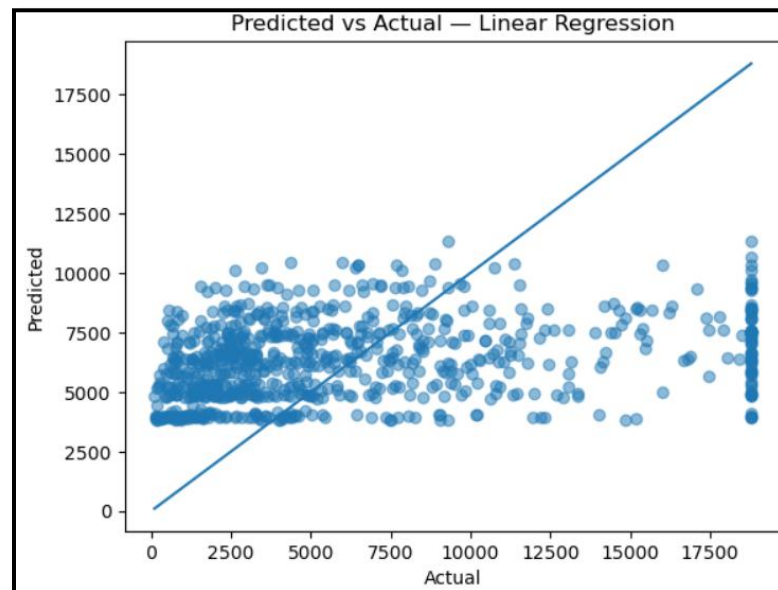
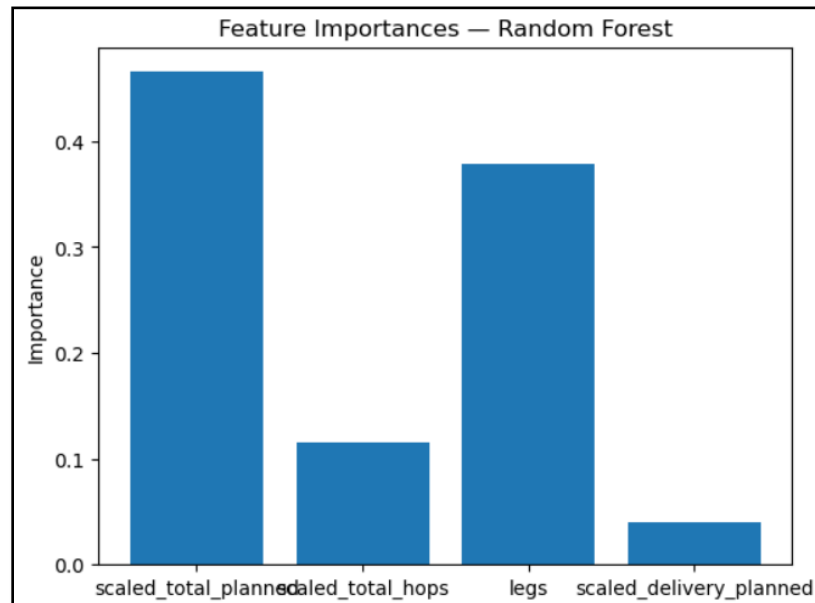


Figure 9. Predicted vs Actual (Random Forest)



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Figure 10. Feature Importance Plot (Random Forest)



Classification Models

- The delays were defined as On-Time, Moderate Delay, and Severe Delay.
- Decision tree classifiers and logistic regression were trained.
- The decision trees performed better than logistic regression on the identification of severe delays; logistic regression provided greater interpretability.

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Figure 11.1 Confusion Matrix (Logistic Regression)

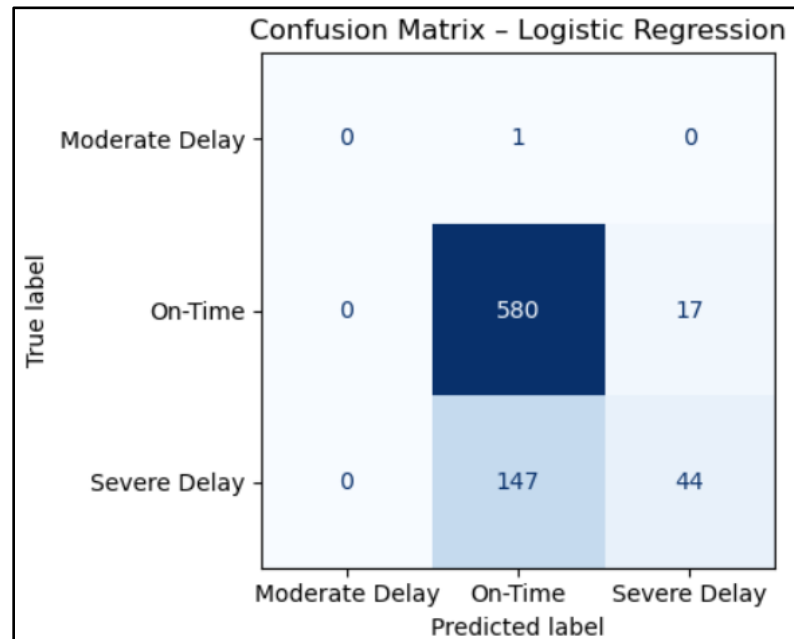
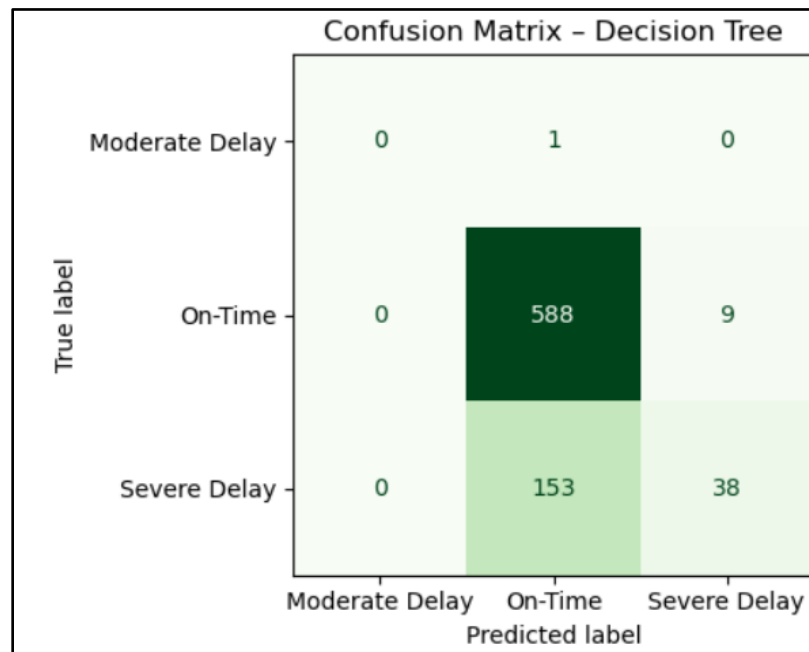
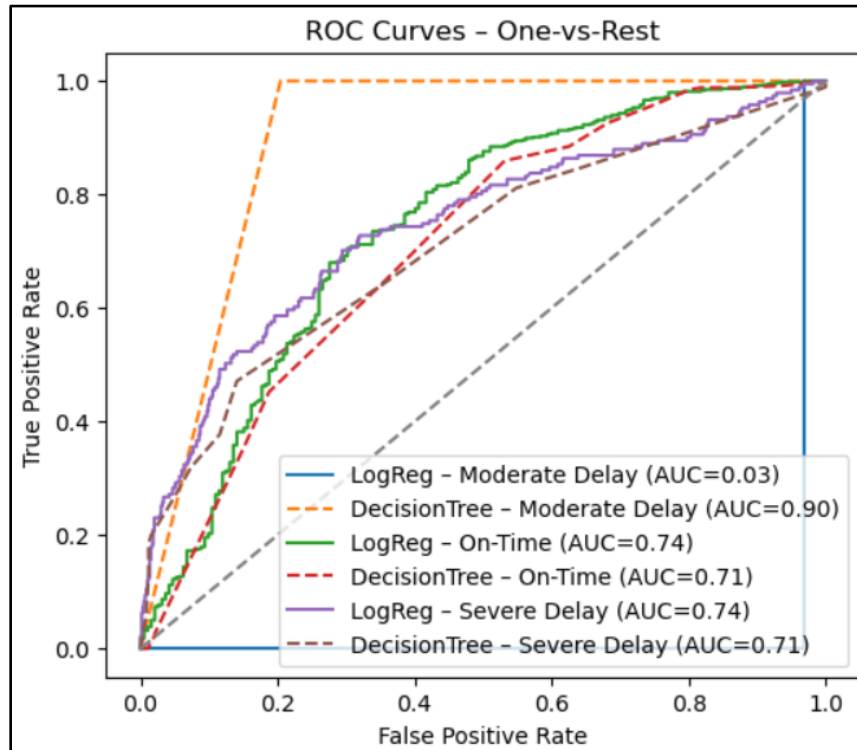


Figure 11.2 Confusion Matrix (Decision Tree)



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Figure 12. ROC Curve for Logistic Regression vs Decision Tree



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Model Evaluation

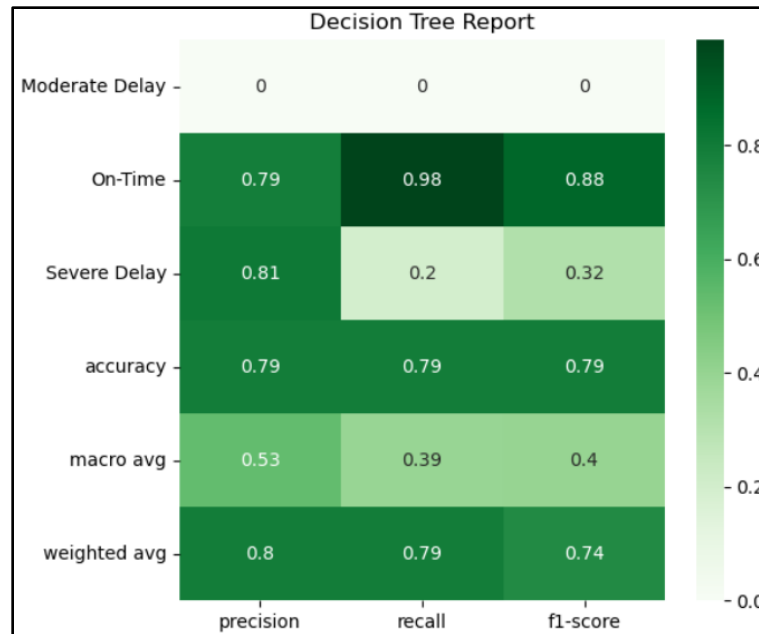
- Regression models: Assessed based on R2, RMSE, and MAE; Random Forest had the best error.
- Classification models: Accuracy, precision, recall, and F1-score were used to evaluate them; decision trees were higher in recalling severe delays.

Figure 13. Logistic Regression Report

Logistic Regression Report			
Moderate Delay -	0	0	0
On-Time -	0.8	0.97	0.88
Severe Delay -	0.72	0.23	0.35
accuracy -	0.79	0.79	0.79
macro avg -	0.51	0.4	0.41
weighted avg -	0.78	0.79	0.75
	precision	recall	f1-score

Figure 14. Decision Tree Report

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Business Impact and Conclusion

The predictive models that are generated are actionable, in the sense that they give the expected delay on transportation of cargos. This helps logistics firms manage schedules proactively, allocate resources effectively and minimize costs related to late deliveries. Predictive insights are also used to enhance customer satisfaction by informing customers of delays on time and supply chain resilience.

Recommendations

- Embed predictive analytics models directly in operational dashboards to monitor in real time. Constantly revise models with new data to aid accuracy.
- It is possible to add external forces (weather, traffic, flight disruptions) to add to predictive knowledge.
- With such recommendations in place, data-driven decision-making can help the logistics organizations optimize their operations to maximize their competitiveness.